

ESE 499 Undergraduate Research –Summer 2026

[Theoretical Review]

Technical Background of SDR Systems: Analysis of PySDR Chapters 1-4

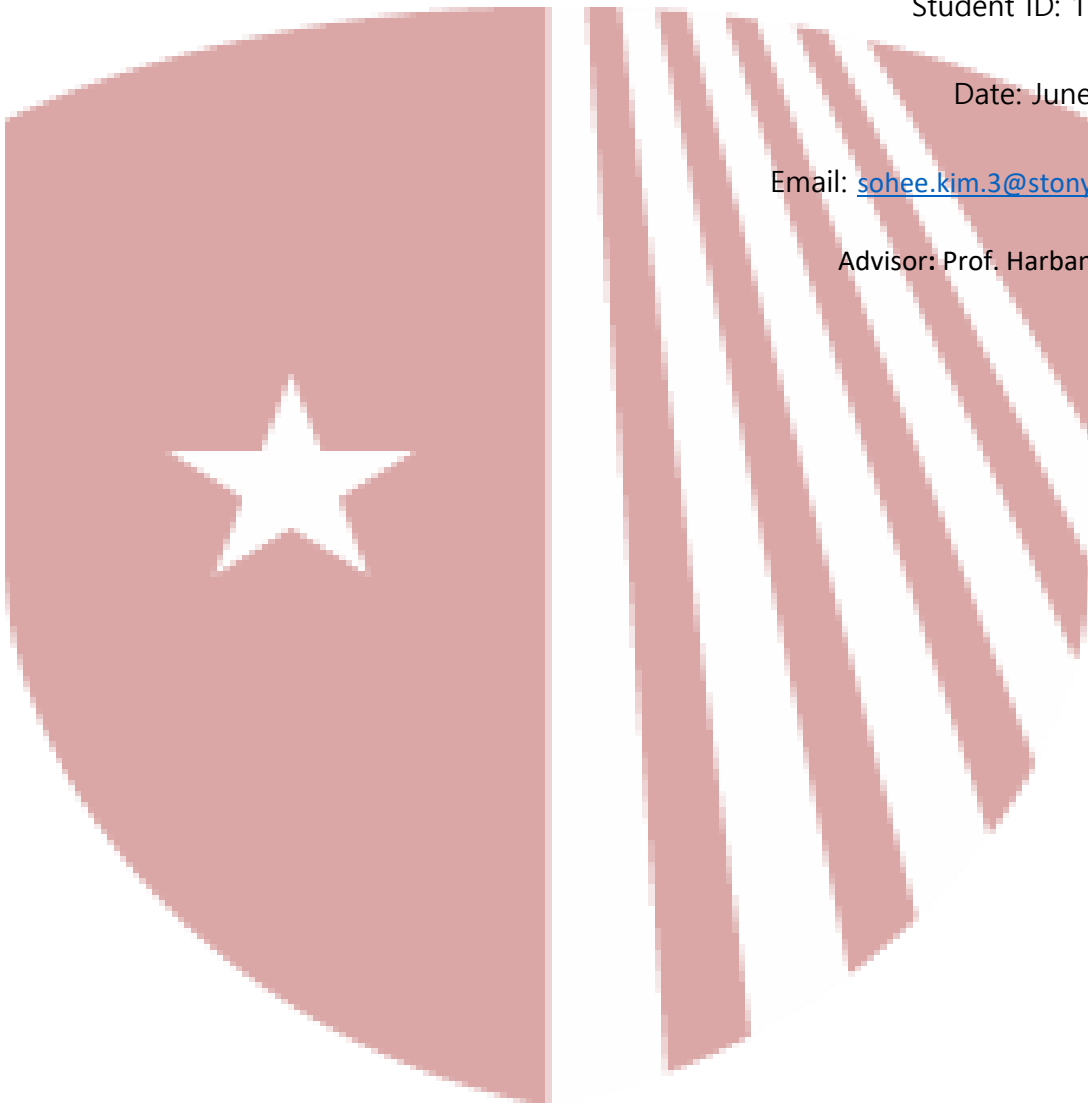
Name: Sohee Kim

Student ID: 117643907

Date: June 11, 2026

Email: sohee.kim.3@stonybrook.edu

Advisor: Prof. Harbans Dhadwal



1. Introduction

This report systemically reviews and analyzes the core theoretical foundations of Digital Signal Processing (DSP) and Software Defined Radio (SDR) architectures, specifically covering the transition from hardware-defined configurations to software-centric IQ processing and digital modulation frameworks.

2. Chapter-by-Chapter Analysis

- Chapter 1: Introduction to SDR and DSP Architecture

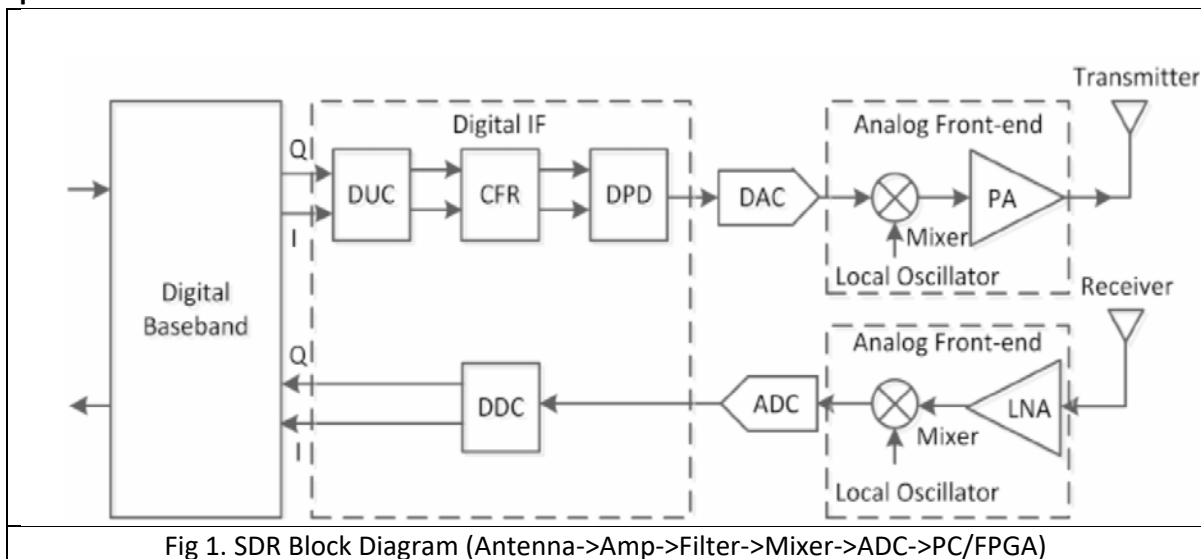


Fig 1. SDR Block Diagram (Antenna->Amp->Filter->Mixer->ADC->PC/FPGA)

Traditional hardware radios rely on fixed analog components, restricting their operational flexibility. Modern SDR architectures circumvent this limitation by digitizing the received Radio Frequency (RF) spectrum at the earliest possible stage using a high-speed Analog-to-Digital Converter (ADC). This structural shift allows digital signal processing algorithms such as filtering, down-conversion, and demodulation to be executed entirely in software on a general-purpose host processor.

- Chapter 2: Frequency Domain Analysis

While continuous-time domain waveforms capture raw voltage variations, they mask critical spectral traits. Decomposing composite signals into their orthogonal frequency components via the Fourier Transform provides vital engineering insights into signal periodicities and occupied bandwidth.

- Continuous Fourier Transform (CFT):

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

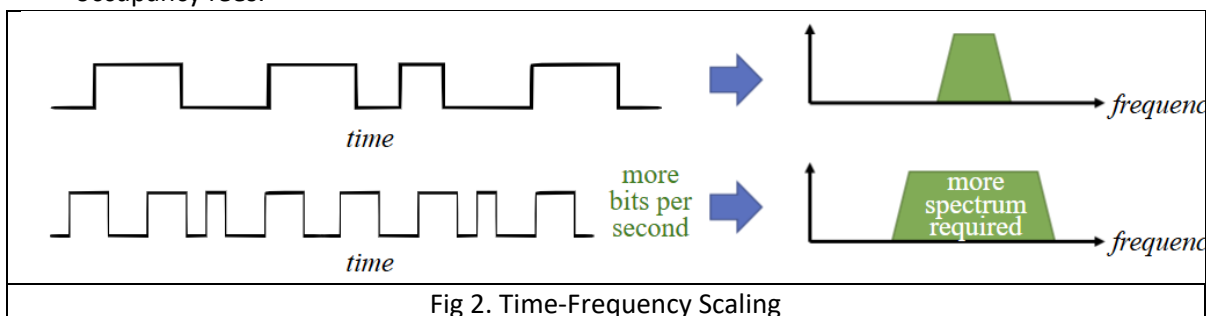
- Discrete Fourier Transform (DFT):

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi}{N}kn}$$

Fundamental Fourier Theorems

1. Linearity: $\mathcal{F}\{ax(t) + by(t)\} = aX(f) + bY(f)$. Enables simultaneous processing of frequency-multiplexed signals without mutual intermodulation distortion.

2. Frequency Shifting: $\mathcal{F}\{e^{j2\pi f_0 t} x(t)\} = X(f - f_0)$. Forms the mathematical foundation for heterodyning, where baseband signals are up-converted to RF carriers or down-converted back to baseband.
3. Time Scaling: $\mathcal{F}\{x(at)\} = \frac{1}{|a|} X\left(\frac{f}{a}\right)$. Establishes an inverse relationship between time duration and spectral footprint. Compressing a signal in time ($a > 1$) to increase data transmission rates inherently widens the required RF transmission bandwidth ($\frac{f}{a}$), leading to higher spectral occupancy fees.

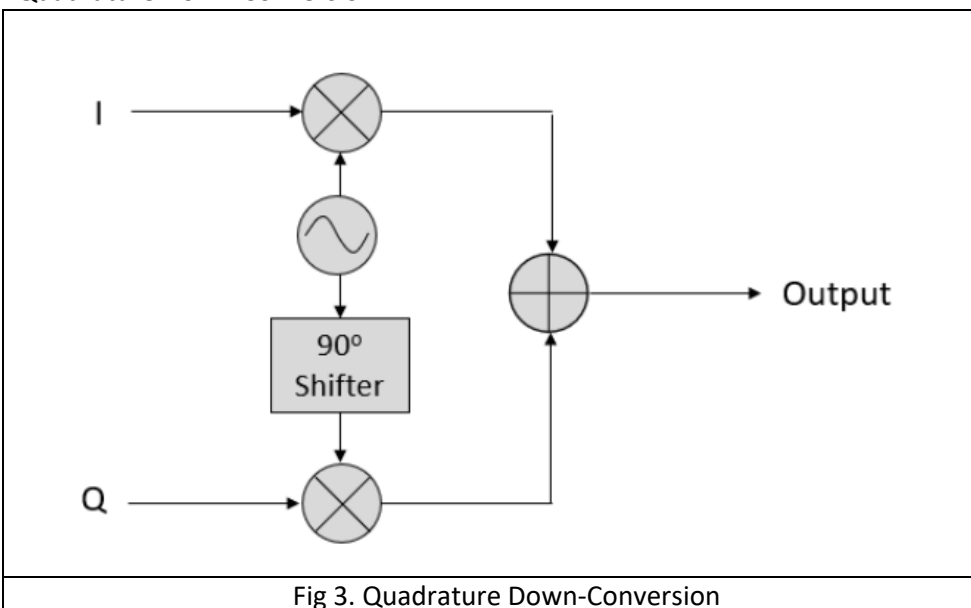


- Chapter 3: Complex In-Phase and Quadrature (IQ) Sampling

Real-valued sampling cannot differentiate between positive and negative frequencies, causing spectral folding (aliasing) around 0 Hz during down-conversion. Splitting the RF signal into two orthogonal components separated by a 90° phase shift—In-phase (I) and Quadrature (Q)—allows the signal to be processed as a complex mathematical entity:

$$x(t) = I(t) + jQ(t)$$

Quadrature Down-Conversion



The incoming signal is mixed with local oscillator (LO) signals in parallel paths:

1. Channel: Mixed with $\cos(2\pi f_c t)$
2. Q-Channel: Mixed with $-\sin(2\pi f_c t)$

This complex vector representation preserves both the instantaneous magnitude and phase:

$$\text{Magnitude } A = \sqrt{I^2 + Q^2}, \quad \text{Phase } \theta = \tan^{-1}\left(\frac{Q}{I}\right)$$

While Nyquist requires $f_s > 2f_{max}$ for real signals, complex IQ sampling isolates the positive and negative spectrums, lowering the minimum sampling rate requirement to the absolute bandwidth of interest ($f_s \geq B$) and optimizing ADC computational throughput.

- Chapter 4: Digital Modulation Schemes and Constellation Space

Digital modulation maps blocks of binary streams onto discrete coordinates within the complex IQ plane, known as symbols.

- Amplitude Shift Keying (ASK): Alters only the magnitude of the complex vector; highly susceptible to noise.
- Phase Shift Keying (PSK): Keeps magnitude constant while shifting the phase angle (θ). Binary PSK (BPSK) maps 1 bit/symbol, whereas Quadrature PSK (QPSK) maps 2 bits/symbol.
- Quadrature Amplitude Modulation (QAM): Simultaneously modulates both amplitude and phase. High-order arrays like 16-QAM or 64-QAM exponentially improve spectral efficiency by packing $\log_2(M)$ bits into an individual symbol.

Plotting received IQ samples yields a Constellation Diagram. In real-world channels governed by Additive White Gaussian Noise (AWGN), discrete symbol points diffuse into clusters. Excessive noise variance forces these clusters to overlap across decision boundaries, inducing symbol misidentification and Bit Error Rate (BER) degradation.

3. Conclusion

Analyzing Chapters 1 through 4 bridges the gap between abstract communication mathematics and software-defined hardware implementation. Understanding Fourier theorems explains band allocation, IQ sampling clarifies how phase is preserved digitally, and constellation analysis highlights how channel degradation limits real-world data rates. This integrated knowledge forms the prerequisite framework for advanced SDR engineering and hardware-in-the-loop experimentation.